

Unit 3: Cloud Virtualization technology

3.1 Overview of Virtualization techniques

- Virtualization is the "creation of a virtual (rather than actual) version of something, such as a server, a desktop, a storage device, an operating system or network resources".
- In other words, Virtualization is a technique which allows to **share a single physical instance** of a resource or an application among multiple customers and organizations.
- It does so by **assigning a logical name** to a physical resource and providing a **pointer to that physical resource** on demand.
- Creating a virtual machine over existing operating system and hardware is referred as Hardware Virtualization. Virtual Machines provide an environment that is logically separated from the underlying hardware.
- The machine on which the virtual machine is created is known as **host machine** and virtual machine is referred as a **guest machine**. This virtual machine is managed by a software or firmware, which is known as hypervisor.

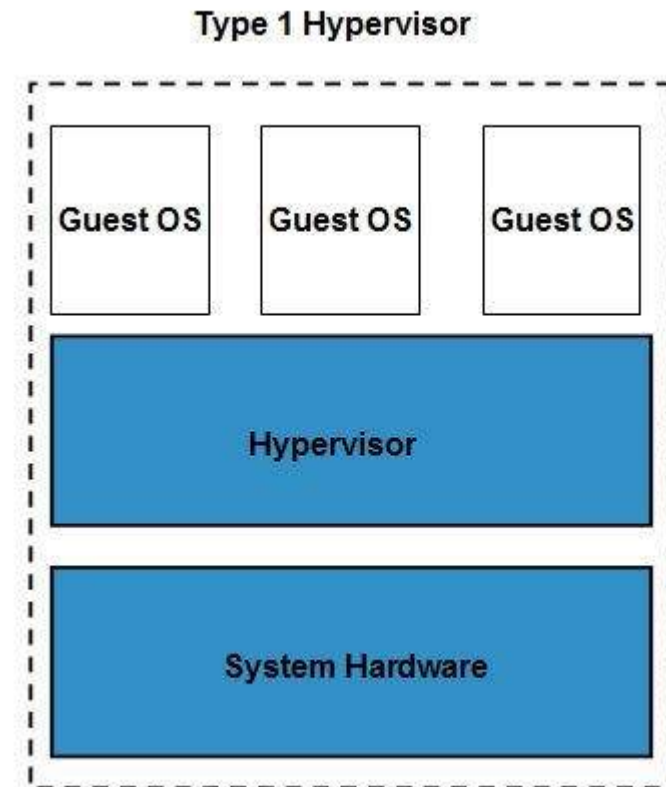
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Hypervisor

The hypervisor is a firmware or low-level program that acts as a Virtual Machine Manager. There are two types of hypervisor:

Type 1 Hypervisor

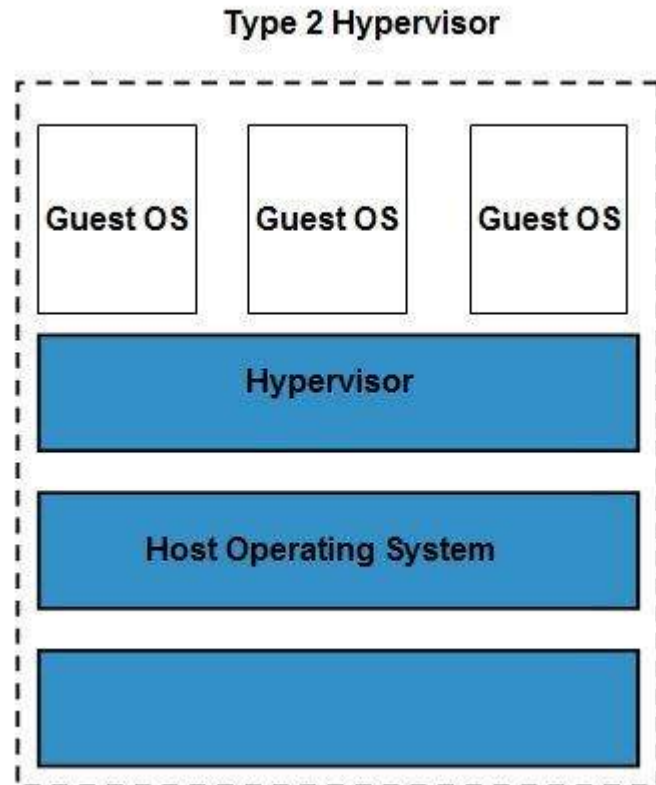
Type 1 hypervisor executes on bare system. LynxSecure, RTS Hypervisor, Oracle VM, Sun xVM Server, VirtualLogic VLX are examples of Type 1 hypervisor. The following diagram shows the Type 1 hypervisor.



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Type 2 Hypervisor

Type 2 hypervisor is a software interface that emulates the devices with which a system normally interacts. Containers, KVM, Microsoft Hyper V, VMWare Fusion, Virtual Server 2005 R2, Windows Virtual PC and VMWare workstation 6.0 are examples of Type 2 hypervisor. The following diagram shows the Type 2 hypervisor.



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3.2 Types of Virtualizations

1. Hardware Virtualization.

- When the virtual machine software or virtual machine manager (*VMM*) is *directly installed on the hardware system* is known as hardware virtualization.
- The main job of hypervisor is to control and monitoring the processor, memory and other hardware resources. After virtualization of hardware system we can install different operating system on it and run different applications on those OS.

Usage:

Hardware virtualization is mainly done for the server platforms, because controlling virtual machines is much easier than controlling a physical server.

2. Operating system Virtualization.

When the virtual machine software or virtual machine manager (VMM) is installed on the Host operating system instead of directly on the hardware system is known as operating system virtualization.

Usage:

Operating System Virtualization is mainly used for testing the applications on different platforms of OS.

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3.2 Types of Virtualizations (contd..)

3. Server Virtualization.

When the virtual machine software or virtual machine manager (*VMM*) is *directly installed on the Server system* is known as server virtualization.

Usage:

Server virtualization is done because a single physical server can be divided into multiple servers on the demand basis and for balancing the load.

4. Storage Virtualization.

Storage virtualization is the process of grouping the physical storage from multiple network storage devices so that it looks like a single storage device.

Storage virtualization is also implemented by using software applications.

Usage:

Storage virtualization is mainly done for back-up and recovery purposes.

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3.3 Implementation Levels of Virtualization Structures

Virtualization structures encompass different implementation levels that offer varying degrees of abstraction between the physical hardware and the virtualized environments running on them. These levels provide different approaches to create and manage virtualized systems. Here are the primary implementation levels of virtualization structures:

1. Full Virtualization:

- In full virtualization, a hypervisor (also known as a Virtual Machine Monitor - VMM) is employed to abstract and manage the physical hardware resources.
- Multiple guest operating systems (OSes) can run simultaneously on the same physical machine, each operating in its isolated virtual machine.
- The hypervisor intercepts and manages hardware calls made by the guest OSes, providing a layer of abstraction and allowing diverse OSes to run without modification.

2. Para-Virtualization:

- Unlike full virtualization, para-virtualization requires modifications to the guest operating systems.
- Guest OSes are aware of the virtual environment and interact with the hypervisor through specialized interfaces or APIs.
- This approach generally offers better performance compared to full virtualization as it eliminates some overhead by having the guest OSes cooperate with the hypervisor.

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3.3 Implementation Levels of Virtualization Structures

3. Hardware-Assisted Virtualization:

- This level of virtualization takes advantage of hardware extensions present in modern CPUs (such as Intel VT-x or AMD-V) to enhance virtualization capabilities.
- Hardware extensions allow the hypervisor to efficiently manage and allocate hardware resources, reducing the overhead associated with software-based emulation.
- It improves the performance of virtualized systems by enabling direct execution of certain privileged instructions.

4. Operating System-Level Virtualization (Containerization):

- Operating system-level virtualization or containerization enables multiple isolated user-space instances, known as containers, to run on a single OS kernel.
- Containers share the host OS's kernel, libraries, and resources, providing lightweight and efficient virtualization for applications.
- Technologies like Docker, Kubernetes, and LXC (Linux Containers) utilize this level of virtualization for rapid deployment and scalability.

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3.3 Implementation Levels of Virtualization Structures

5. Application-Level Virtualization:

- At this level, virtualization occurs at the application layer, abstracting and encapsulating individual applications or services.
- Applications are packaged with their dependencies and runtime environment, ensuring consistency and portability across different computing environments.
- Technologies such as Java Virtual Machine (JVM) and Microsoft's .NET framework provide application-level virtualization by executing code in managed runtime environments.

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3.4 Virtualization benefits

- 1. Resource Utilization and Efficiency:** Virtualization enables the consolidation of multiple virtual machines (VMs) or containers onto a single physical server, maximizing hardware utilization. By sharing resources dynamically among VMs or containers, it optimizes server capacity and reduces underutilization of hardware.
- 2. Cost Savings:** Through consolidation, organizations can reduce the number of physical servers needed, leading to lower hardware, maintenance, and operational costs. Additionally, virtualization allows for better energy efficiency by running fewer physical machines.
- 3. Improved Scalability and Flexibility:** Virtualized environments can easily scale up or down based on demand. Administrators can allocate resources dynamically to VMs or containers as needed, enabling rapid provisioning and scaling of applications and services.
- 4. Isolation and Security:** Virtualization provides a layer of isolation between different VMs or containers, enhancing security by containing potential security breaches within one instance and preventing them from affecting others. It also facilitates easier backup and recovery processes.

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3.4 Virtualization benefits

5. Simplified Management and Administration: Centralized management tools for virtualized environments simplify administrative tasks such as provisioning, monitoring, and maintenance. Automated provisioning and deployment streamline the management of IT resources.

6. High Availability and Disaster Recovery: Virtualization solutions offer features like live migration and high availability, enabling seamless movement of VMs between physical servers and minimizing downtime during maintenance or hardware failures. This aids in disaster recovery and business continuity planning.

7. Testing and Development Efficiency: Virtualization allows for the creation of isolated test environments, enabling developers to test applications in various configurations without affecting the production environment. It accelerates the development lifecycle and enhances testing efficiency.

8. Compatibility and Legacy System Support: Virtualization enables running older or incompatible software and operating systems on modern hardware by providing an abstraction layer. This supports legacy applications that might not be compatible with the latest hardware or operating systems.

9. Cloud Readiness: Virtualization forms the foundation for cloud computing by providing the fundamental abstraction of resources. Cloud services are often built on virtualized environments, allowing for the efficient delivery of resources over the internet.

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3.5 Server virtualization

Server virtualization is a form of virtualization technology where a physical server is abstracted into multiple virtual machines (VMs), allowing multiple operating systems and applications to run independently on a single physical server. This process is managed by a hypervisor, also known as a Virtual Machine Monitor (VMM), which allocates and manages the server's resources among the virtualized environments.

Here are key aspects and components of server virtualization:

1. Hypervisor: The hypervisor is the core component of server virtualization. It creates and manages multiple VMs on a single physical server by abstracting the underlying hardware resources. There are two types of hypervisors:

Type 1 Hypervisor (Bare-Metal): Installed directly on the physical hardware, this hypervisor operates independently of the host operating system. Examples include VMware vSphere/ESXi, Microsoft Hyper-V, and KVM.

Type 2 Hypervisor (Hosted): Installed on top of a host operating system, this hypervisor functions as an application. Examples include VMware Workstation, Oracle VirtualBox, and Parallels Desktop.

2. Virtual Machines (VMs): These are the individual instances created by the hypervisor that behave like independent servers. Each VM runs its own operating system and applications, which are isolated from other VMs on the same physical server.

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3.5 Server virtualization (contd..)

3. Resource Allocation and Management: The hypervisor dynamically allocates physical server resources such as CPU, memory, storage, and network bandwidth among VMs based on their requirements. This allows for efficient utilization of hardware resources and flexible scaling.

4. Live Migration: Some hypervisors offer live migration capabilities, allowing VMs to be moved between physical servers without disrupting services. This feature aids in load balancing, maintenance, and avoiding downtime during hardware failures.

5. Snapshotting and Cloning: Virtualization platforms often provide snapshot and cloning functionalities. Snapshots capture the state of a VM at a specific point in time, allowing for easy restoration if needed. Cloning enables the rapid duplication of VMs for testing or deployment purposes.

6. High Availability and Fault Tolerance: Server virtualization technologies offer features to ensure high availability and fault tolerance. Redundancy, failover mechanisms, and automated recovery processes help maintain system availability and minimize disruptions.

7. Centralized Management: Virtualization platforms typically include centralized management consoles or tools that enable administrators to manage multiple VMs, monitor performance, allocate resources, and configure settings from a single interface.

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3.6 Hypervisor Management Software

Hypervisor management software, also known as Virtual Machine Management (VMM) or Virtualization Management software, is designed to facilitate the administration, configuration, monitoring, and optimization of virtualized environments. This software is essential for effectively managing multiple virtual machines (VMs) or containers running on a hypervisor. Here are some notable hypervisor management software solutions:

- 1. VMware vSphere:** VMware's vSphere is a comprehensive suite of products that includes ESXi hypervisor and vCenter Server for managing virtualized environments. vSphere provides centralized management, high availability, resource optimization, and various features for scalability and security.
- 2. Microsoft Hyper-V Manager:** Part of the Windows Server platform, Hyper-V Manager allows administrators to create, manage, and monitor VMs running on Microsoft's Hyper-V hypervisor. It provides functionalities for configuring virtual networks, snapshots, and storage.
- 3. Citrix Hypervisor (formerly XenServer):** Citrix Hypervisor is an open-source virtualization platform that offers features for server virtualization and cloud infrastructure management. It includes tools for workload balancing, live migration, and disaster recovery.

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3.6 Hypervisor Management Software (contd..)

4. KVM (Kernel-based Virtual Machine): KVM is a Linux-based open-source hypervisor that is integrated into the Linux kernel. Management tools like oVirt and Kimchi provide graphical interfaces for managing KVM-based virtualized environments.

5. Oracle VM VirtualBox: VirtualBox is a free, open-source hypervisor that supports multiple guest operating systems. It offers a user-friendly interface for creating and managing VMs on desktop systems for testing, development, or production use.

6. Proxmox Virtual Environment: Proxmox is an open-source platform that combines KVM-based virtualization and containerization (LXC) within a single management interface. It provides features for VM management, high availability, and backup solutions.

7. OpenStack: OpenStack is an open-source cloud computing platform that manages pools of compute, storage, and networking resources. It includes components like Nova (compute), Cinder (block storage), and Neutron (networking) for managing virtualized infrastructure.

8. Docker Enterprise: Docker Enterprise provides container orchestration and management tools for deploying and managing containerized applications at scale. It includes features for container lifecycle management, security, and clustering.

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3.7 Virtual infrastructure requirements

Creating and maintaining a virtual infrastructure involves several key requirements to ensure its effectiveness, performance, security, and manageability. Here are essential components and considerations for establishing a robust virtual infrastructure:

1. Hardware Resources:

- a. Server Hardware:** High-performance servers with adequate CPU cores, memory (RAM), storage (hard drives or SSDs), and network connectivity are essential. Hardware compatibility with virtualization technologies is crucial.
- b. Networking Infrastructure:** Robust and reliable network components, such as switches, routers, and network interfaces, are necessary to ensure efficient communication between virtualized environments and external networks.

2. Hypervisor or Virtualization Software: Choosing the right hypervisor or virtualization platform (such as VMware vSphere, Microsoft Hyper-V, KVM, Xen, or containerization platforms like Docker or Kubernetes) that suits the organization's needs is critical. Ensure compatibility with hardware and desired features.

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3.7 Virtual infrastructure requirements

3. Storage Infrastructure:

- a. Storage Area Network (SAN) or Network Attached Storage (NAS):** These centralized storage solutions facilitate storage provisioning, data redundancy, and scalability for virtual machines.
- b. Storage Virtualization:** Implementing storage virtualization technologies allows for efficient utilization and management of storage resources.

4. Management and Monitoring Tools:

- a. Hypervisor Management Software:** Utilize tools specific to the chosen hypervisor for provisioning, monitoring, and managing virtual machines, ensuring ease of administration and optimization.
- b. Monitoring Solutions:** Implement monitoring tools to track resource usage, performance metrics, and potential issues within the virtualized environment.

5. Backup and Disaster Recovery:

- a. Backup Solutions:** Employ reliable backup mechanisms to protect data and virtualized workloads, ensuring quick recovery in case of data loss or system failures.
- b. Disaster Recovery Plan:** Develop and implement a robust disaster recovery strategy to maintain business continuity in the event of catastrophic failures or disruptions.

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3.7 Virtual infrastructure requirements

6. Security Measures:

a. Network Security: Implement firewalls, intrusion detection systems, and secure network configurations to protect virtualized environments from external threats.

b. Access Controls: Enforce strict access controls, user authentication mechanisms, and role-based access to prevent unauthorized access to virtual infrastructure resources.

c. Security Updates and Patch Management: Regularly apply security updates and patches to hypervisors, virtual machines, and other components to mitigate vulnerabilities.

7. Scalability and Performance Optimization:

Ensure that the virtual infrastructure can scale to accommodate growing workloads and demands.

Optimize resource allocation and performance tuning for virtual machines to achieve maximum efficiency and responsiveness.

8. Compliance and Regulations:

Ensure that the virtual infrastructure complies with relevant industry standards, regulations, and data protection laws to maintain data integrity, confidentiality, and compliance.

independent computer with its own OS and applications.

•Purpose: Run multiple operating systems on one physical machine.

Virtualization is the process of creating a virtual version of something, such as hardware, software, storage, network, or an operating system, allowing a single physical resource to function as multiple resources. It maximizes efficiency and flexibility by running multiple virtual environments on the same physical system.

Benefits of Virtualization

•Examples:

•Cost-Effective: Reduces the need for physical hardware.

•Efficient Resource Usage: Maximizes utilization of hardware resources.

•Scalability: Easily add or reduce virtual resources as needed.

•Isolation: Provides secure environments for testing and deployment.

•Use Case: Cloud hosting services like AWS or Google Cloud where multiple users run isolated virtual environments on shared hardware.

•Disaster Recovery: Simplifies backup and recovery processes.

•Examples:

•VirtualBox

•VMware Workstation

•Microsoft Hyper-V